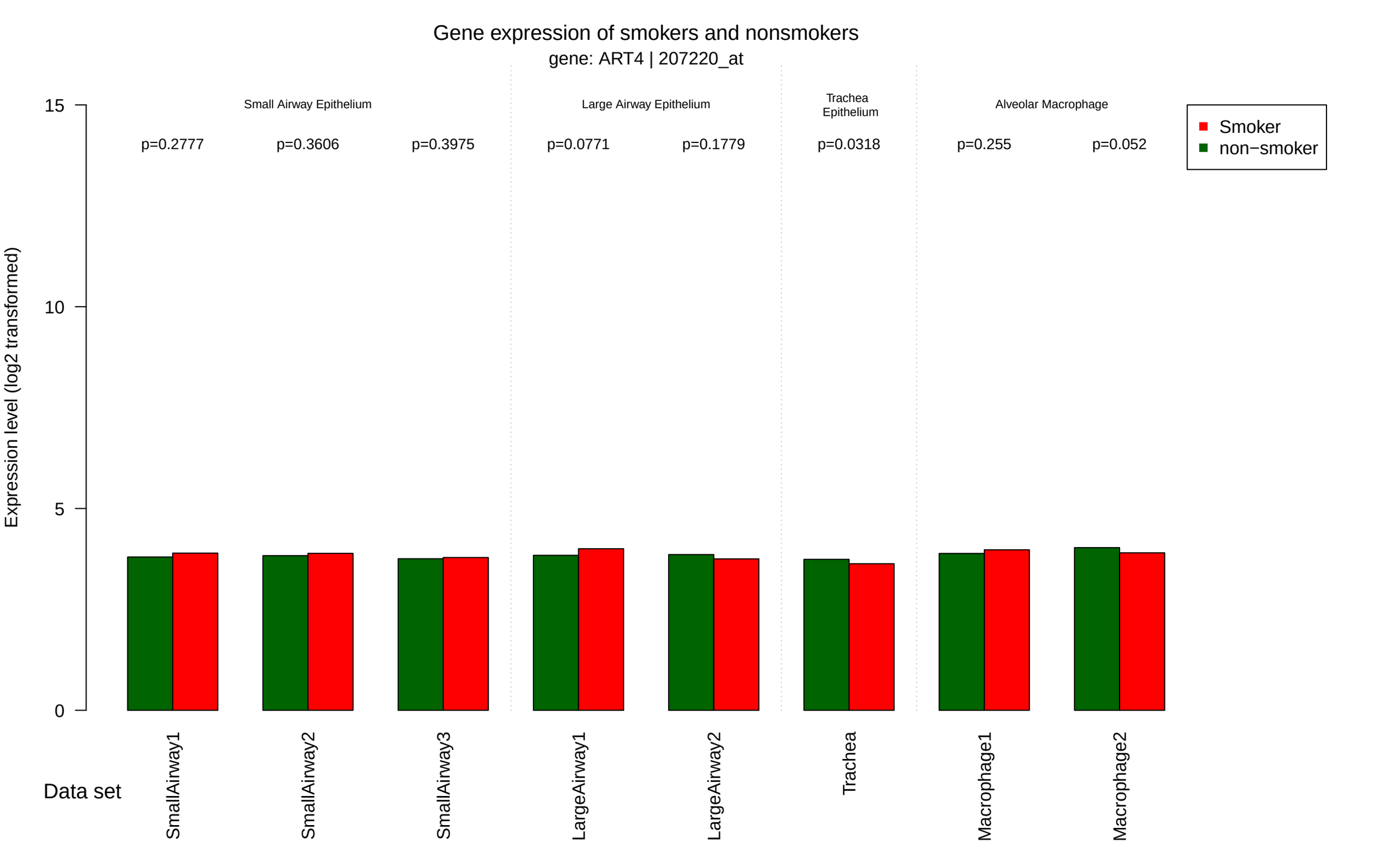
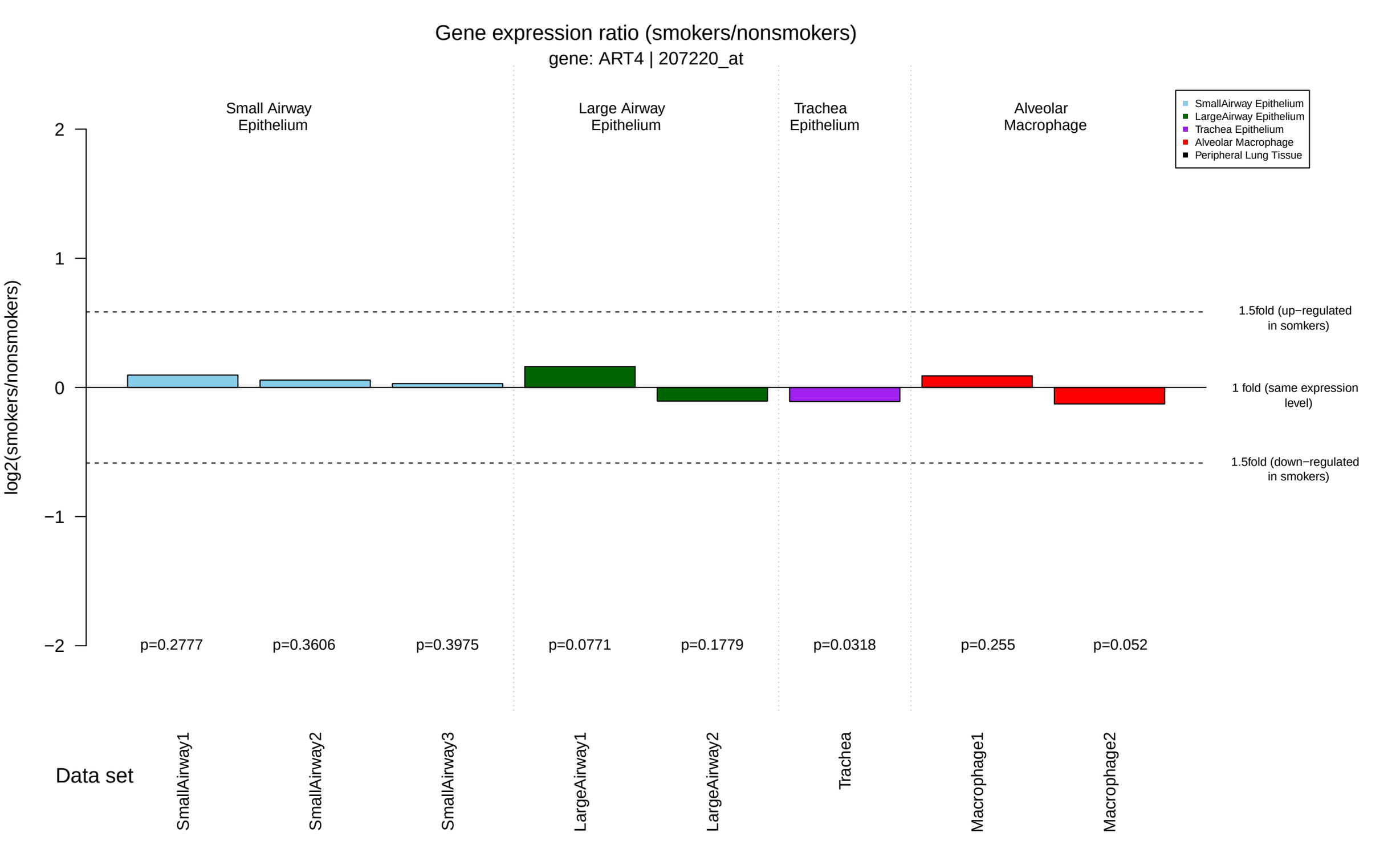
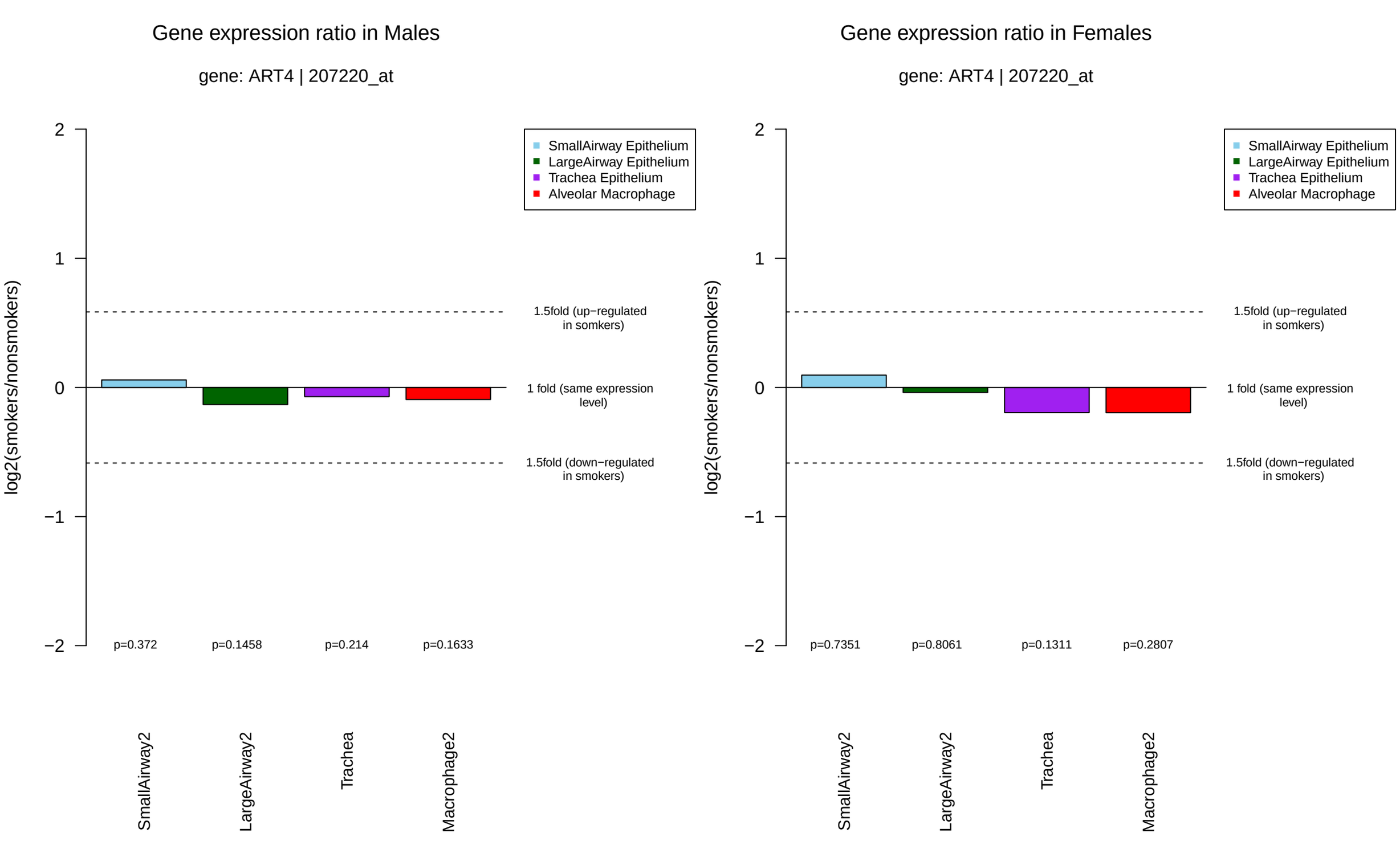
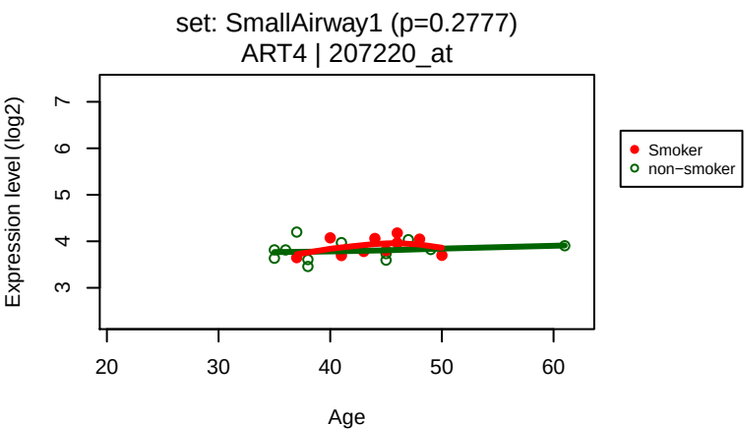
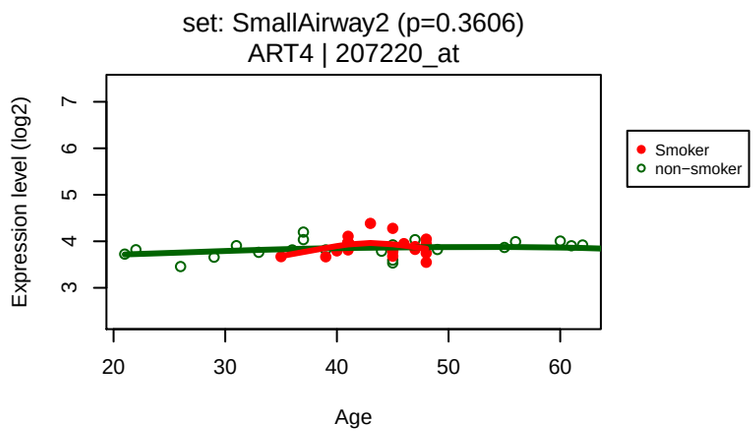
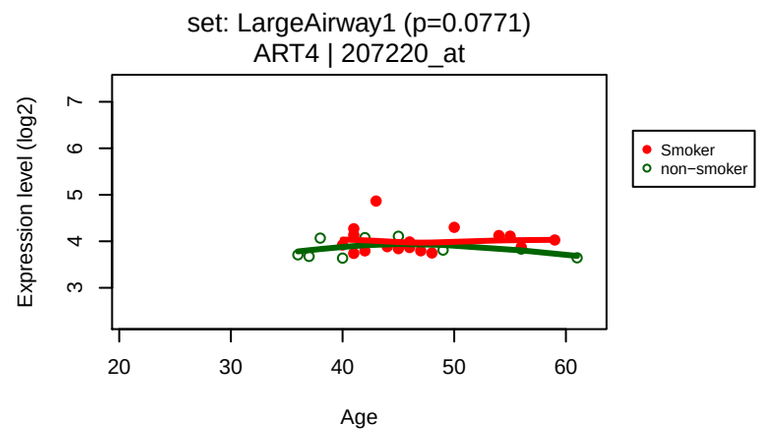

Gene expression of smokers and nonsmokers
gene: ART4 | 207220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2777
p=0.3606
p=0.3975
p=0.0771
p=0.1779
p=0.0318
p=0.255
p=0.052
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

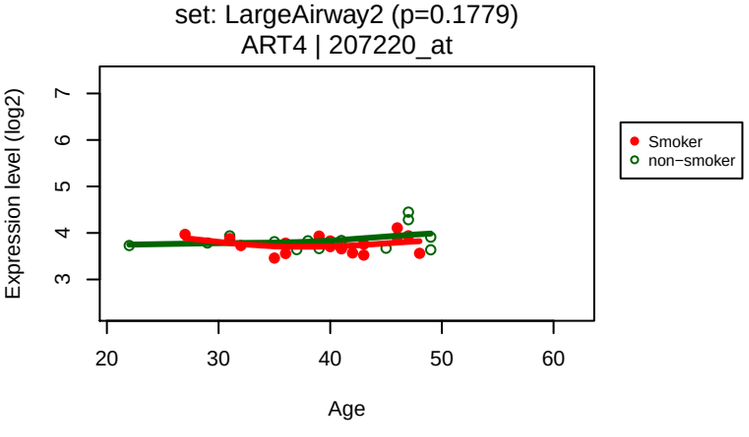

Gene expression ratio (smokers/nonsmokers)
gene: ART4 | 207220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2777
p=0.3606
p=0.3975
p=0.0771
p=0.1779
p=0.0318
p=0.255
p=0.052
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

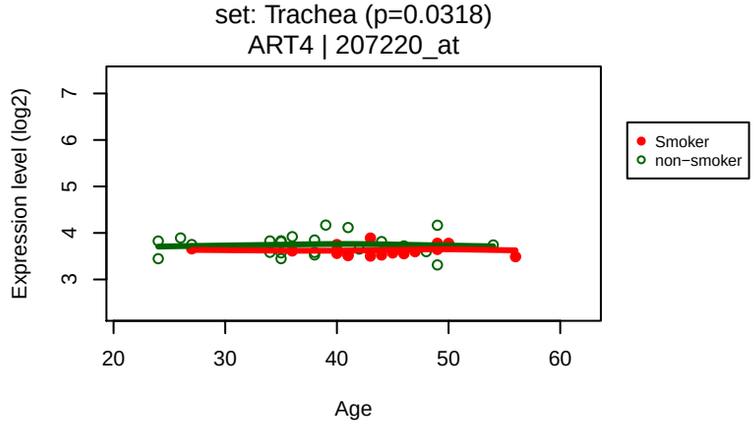

Gene expression ratio in Males
gene: ART4 | 207220_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.372
p=0.1458
p=0.214
p=0.1633
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ART4 | 207220_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7351
p=0.8061
p=0.1311
p=0.2807
SmallAirway2
LargeAirway2
Trachea
Macrophage2

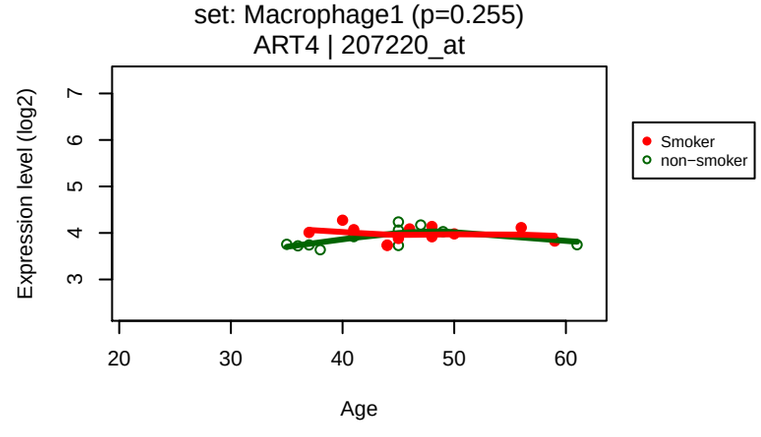

set: SmallAirway1 (p=0.2777)
ART4 | 207220_at
Expression level (log2)
Age
Smoker
non−smoker

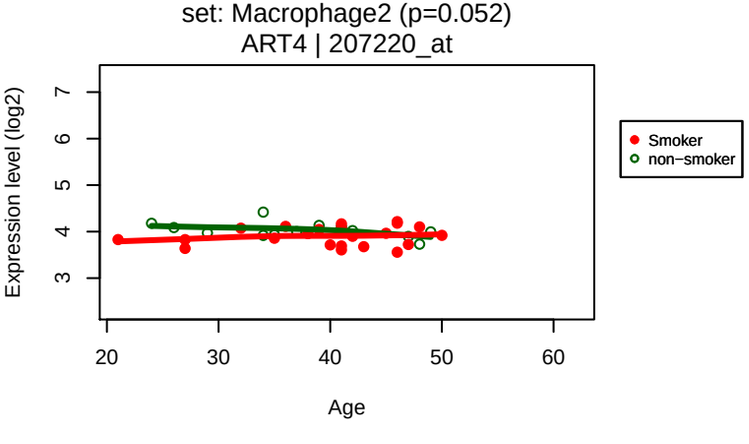

set: SmallAirway2 (p=0.3606)
ART4 | 207220_at
Expression level (log2)
Age
Smoker
non−smoker

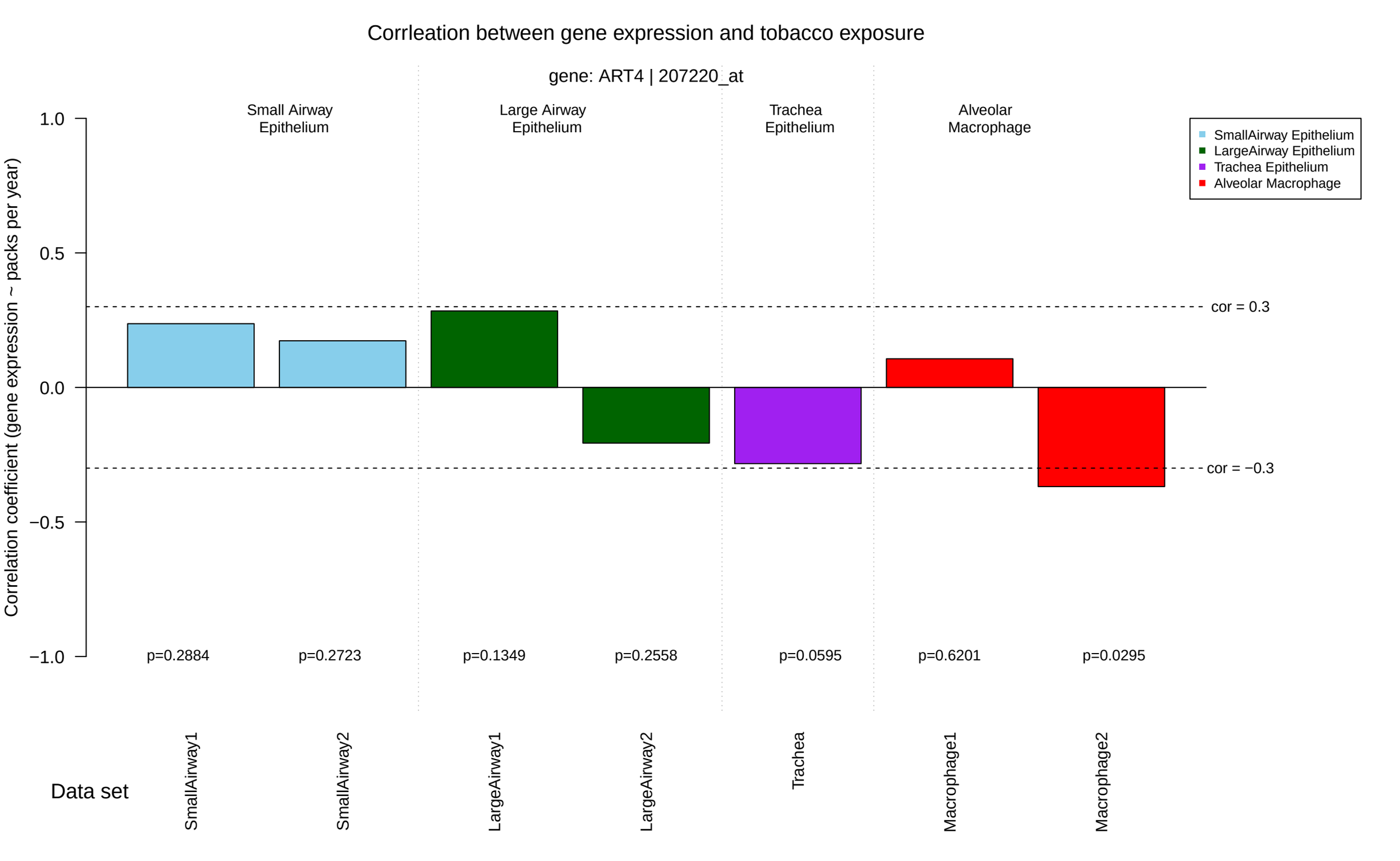

set: LargeAirway1 (p=0.0771)
ART4 | 207220_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1779)
ART4 | 207220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0318)
ART4 | 207220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.255)
ART4 | 207220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.052)
ART4 | 207220_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ART4 | 207220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2884
p=0.2723
p=0.1349
p=0.2558
p=0.0595
p=0.6201
p=0.0295
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2